

I am a postdoctoral assistant professor at the University of Michigan, working in algebra and algebraic geometry.

Specifically, I am interested in: derived categories of projective varieties and quivers; DG and A_∞ algebras; Bridgeland stability; DM stacks, twisted sheaves and Brauer groups; stable sheaves and moduli spaces; Brill–Noether theory; classical low-dimensional algebraic geometry.

Employment & Education

University of Michigan, Postdoctoral Assistant Professor

Sep. 2024–present

Mentor: Alexander Perry

University of Illinois Chicago, Ph.D. Mathematics

2019–2024

Advisor: Izzet Coskun

Nanjing University, B.S. Mathematics

2015–2019

Research

1. *Phantom categories on 3-vertex directed DG quivers*
arXiv:2609.27191; with Yu Shen
2. *Cartan determinants in algebraic geometry*
preprint available upon request; with Yu Shen and Ziyu Zhang
3. *A counterexample to a global-dimension bound for weighted projective lines*
arXiv:2608.23981; with Bochao Kong and Yu Shen
4. *A counterexample to Han’s conjecture*
arXiv:2608.00177, submitted; with Bochao Kong and Yu Shen
5. *A counterexample to DG version of Han’s conjecture*
arXiv:2512.12460, submitted; with Yu Shen
6. *Picard group action on the category of twisted sheaves*
J. Noncommut. Geom. (2026), published online first; with Ting Gong and Yu Shen
7. *A looming of phantoms*
Adv. Math. 499 (2026), 111046; with Kimoi Kemboi, Daniel Krashen, Tianle Liu, Eoin Mackall, Svetlana Makarova, Alexander Perry, Antonios-Alexandros Robotis, and Sridhar Venkatesh
8. *Geometric phantom categories do not admit Noetherian t -structures*
arXiv:2503.02052, submitted
9. *Stringy Chow rings and weighted blow-ups*
Int. Math. Res. Not. 2026, no. 3, rnaf374; with Qiangru Kuang, Rachel Webb, and Weihong Xu
10. *Morita theory on root gerbes*
arXiv:2409.20317, submitted; with Yu Shen
11. *Stability conditions on surface root stacks*
arXiv:2407.19104, submitted; with Yu Shen
12. *Cones of effective cycles on blow ups of projective spaces along rational curves*
arXiv:2308.13964, submitted; with Benjamin Gould
13. *Spherical vector bundles on nodal $K3$ surfaces*
arXiv:2307.00091, submitted
14. *Nonexistence of exceptional bundles on $\mathbb{P}^3$ with maximal possible ranks*
J. Algebra 695 (2026), 283–310
15. *The cohomology of spherical vector bundles on $K3$ surfaces*
arXiv:2210.11030; preprint being polished
16. *Higher rank Brill–Noether theory on $\mathbb{P}^2$*
Int. Math. Res. Not. 2023, no. 24, 22096–22137; with Benjamin Gould and Woohyung Lee

Awards

UIC Dean’s Scholar Fellowship

Fall 2023

Organized Events

<i>Midwest Algebraic Geometry Graduate Student Conference 2022</i>	Summer 2022
Co-organized with Sixuan Lou, Shravan Patankar, Benjamin Tighe, and Junyan Zhao	
<i>Derived Category Student Seminar</i>	Spring 2022
Co-organized with Benjamin Gould	
<i>UIC Graduate Algebraic Geometry Seminar</i>	Spring 2022
Co-organized with Kuang Yu Wu	
<i>Gromov–Witten Theory Student Seminar</i>	Winter 2021
<i>UIC Graduate Algebraic Geometry Seminar</i>	Fall 2021
Co-organized with Kuang Yu Wu	

Mentoring

<i>Directed Reading Program on quiver representations</i>	Spring 2026
Mentees: Yotam Chopra and Xiangnong Wu	
<i>Directed Reading Program on algebraic geometry</i>	Spring 2020
Mentee: Neelima Borade	

Teaching

University of Michigan	2024–2026
Calculus II (Fall 2024), Linear Algebra (Spring 2025), Calculus I (Fall 2025), Multivariable Calculus (Spring 2026), Calculus II (Fall 2026)	
University of Illinois Chicago	2019–2023
Mathematical Reasoning (Fall 2019), Elementary Linear Algebra (Spring 2020), Calculus I (Fall 2020), Calculus I (Spring 2021), Calculus I (Fall 2021), Calculus II (Spring 2022), Precalculus (Summer 2022), Mathematical Reasoning (Fall 2022), Calculus I (Spring 2023), Calculus II (Summer 2023), Calculus II (Fall 2023)	